

Stochastic Solar Resource Modelling & BESS Capacity Optimisation

A Technical Framework for Engineering Mathematics & Applied Statistics

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1. Executive Summary

Deterministic renewable asset models rely heavily on static, historical "average day" parameters, which mask underlying resource volatility. In real-world power grid operations, meteorological fluctuations introduce substantial operational risk. This project develops a clean mathematical and statistical framework to model solar irradiance volatility and optimise Battery Energy Storage System (BESS) sizing. By deploying continuous probability distributions, joint probability matrices, and differential sensitivity calculus, the model computes the exact financial inflection point where expanding storage capacity balances out grid asset risk, completely independent of external software libraries.

2. Statistical Resource Modelling

2.1 Clearness Index Formulation

To isolate the variable impact of passing cloud cover from predictable diurnal movements, the hourly Global Horizontal Irradiance (GHI) is transformed into a dimensionless parameter known as the Clearness Index (K_t):

$$K_t = \text{GHI} / \text{GHI}_{\text{clear-sky}}$$

2.2 Beta Probability Density Function

Because solar irradiance constraints are strictly bounded between 0 (complete darkness) and 1 (perfectly clear clear-sky conditions), standard Gaussian (normal) distributions are physically inappropriate. The resource is instead analytically modeled using a continuous Beta Probability Density Function (PDF), defined by Gamma (Γ) functions:

$$f(K_t; \alpha, \beta) = [\Gamma(\alpha + \beta) / (\Gamma(\alpha) \cdot \Gamma(\beta))] \cdot K_t^{\alpha-1} \cdot (1 - K_t)^{\beta-1}$$

Valid for boundary conditions: $0 \leq K_t \leq 1$

The required shape parameters (α, β) are calculated directly from the statistical mean (μ) and variance (σ^2) of the raw field dataset using the following analytical transformations:

$$\begin{aligned} \alpha &= \mu \cdot [(\mu \cdot (1 - \mu) / \sigma^2) - 1] \\ \beta &= (1 - \mu) \cdot [(\mu \cdot (1 - \mu) / \sigma^2) - 1] \end{aligned}$$

3. Mathematical System Optimisation

3.1 Economic Objective Function

Determining the optimal capital scale of a BESS asset requires balancing structural procurement costs against the financial value of recovered curtailment. The model establishes an economic objective function designed to maximise project Net Present Value (NPV):

$$\max_{[E_{\text{BESS}}, P_{\text{BESS}}]} \text{NPV} = \sum_{y=1..N} [(\Delta R_y(E_{\text{BESS}}, P_{\text{BESS}}) - \text{OPEX}_y) / (1 + r)^y] - \text{CAPEX}(E_{\text{BESS}}, P_{\text{BESS}})$$

3.2 Technical Operational Boundary Conditions

The multi-variable optimisation problem is strictly governed by physical network envelopes:

- **Power Capacity Envelope:** $0 \leq P_{\text{charge}}(t) \leq P_{\text{BESS}}$ and $0 \leq P_{\text{discharge}}(t) \leq P_{\text{BESS}}$
- **Energy Storage Bounds:** $\text{SOC}_{\text{min}} \cdot E_{\text{BESS}} \leq E_{\text{stored}}(t) \leq \text{SOC}_{\text{max}} \cdot E_{\text{BESS}}$
- **Dynamic Energy Continuity:** $E_{\text{stored}}(t) = E_{\text{stored}}(t-1) + [P_{\text{charge}}(t) \cdot \eta_c] \Delta t - [P_{\text{discharge}}(t) / \eta_d] \Delta t$

4. Analytical Sensitivity & Risk Evaluation

4.1 Partial Derivatives for Marginal Returns

To isolate the exact physical inflection point of diminishing capital returns, the system resolves the partial derivative of total annual Curtailment Reduction (ΔC) relative to incremental step changes in Battery Energy Capacity (E_{BESS}):

$$\partial \Delta C / \partial E_{\text{BESS}} = f(K_t, \text{Load}, \eta_{\text{round-trip}})$$

As asset capacity expands toward infinity, the marginal recovery rate asymptotically approaches zero, proving that over-sizing creates severe, non-recoverable capital under-utilisation:

$$\lim_{(E_{\text{BESS}} \rightarrow \infty)} [\partial \Delta C / \partial E_{\text{BESS}}] = 0$$

4.2 Joint Probability of Loss of Power Supply (LPS)

The engineering reliability risk profile is mapped as the joint probability that the solar generation state (S) falls short of the consumer load demand (L) concurrently while the internal battery energy storage state (B) is completely depleted at its minimum SOC floor:

$$P(\text{LPS}) = P(S(t) < L(t) \cap B(t) = B_{\text{min}})$$

By establishing a stochastic joint-probability matrix across 1,000 distinct weather variations, the model demonstrates that a 240 MWh system bounds P(LPS) below 0.015 (1.5%), creating the technical safety cushion required by grid network infrastructure operators.

BESS Capacity (MWh)	Annual Curtailment	Renewable Fraction	System Reliability Risk [P(LPS)]
0 MWh (No BESS)	38.4%	44.1%	14.2%

100 MWh BESS	24.1%	61.8%	6.5%
240 MWh (Optimal)	11.2%	78.4%	<1.5%
400 MWh BESS	1.2%	89.5%	<0.1%